# Databricks ML Associate

## High-Yield Next 100 (Set 4)

Practice Exam Deck

# How to Use This Deck

100 more questions (Q1–Q100), complementing Sets 1 to 3, on topics that commonly appear on the Databricks Machine Learning Associate exam.

Each question slide has four options (A–D); the next slide gives the correct answer and why the others are wrong.

Topic order: Spark ML transformers; MLflow tracking extras; Delta and streaming; statistics and EDA; ML theory; distributed ML; Hyperopt extras; evaluation; deployment and monitoring; workspace.

# Exam Domains Covered

**Databricks Machine Learning:** runtime, AutoML, MLflow, Unity Catalog registry, Feature Engineering client.

**Data Processing:** EDA, imputation, encoding, Spark ML transformers, pandas API on Spark and UDFs.

**Model Development:** Hyperopt, CrossValidator, metrics, bias/variance, leakage.

**Model Deployment:** batch, streaming, real-time serving, monitoring.

# Question 1

In Spark 3 and later, `OneHotEncoder` is which type of Spark ML object?

A. A model registry client
B. A plain transformer only
C. An estimator that must be fit before `transform`
D. An evaluator

# Answer 1

**Correct: C** — it learns the category sizes during `fit`.

**Wrong:**

A: unrelated.

B: it needs to be fit in Spark 3+.

D: evaluators compute metrics.

# Question 2

What does `OneHotEncoder(dropLast=True)` (the default) do for a feature with 5 categories?

A. Produces a vector of length 5
B. Produces a vector of length 1
C. Drops the entire column
D. Produces a vector of length 4, dropping the last category

# Answer 2

**Correct: D** — the last category is the implicit baseline.

**Wrong:**

A: that is `dropLast=False`.

B: not the behavior.

C: not the behavior.

# Question 3

By default, which category does `StringIndexer` assign index 0?

A. The alphabetically first label
B. The most frequent label
C. A random label
D. The rarest label

# Answer 3

**Correct:** **B** — the default order is by descending frequency.

**Wrong:**

A: not the default.

C: not the default.

D: not the default.

# Question 4

Which Spark ML class converts predicted numeric indices back to original string labels?

A. `VectorSlicer`

B. `LabelDecoder`

C. `IndexToString`

D. `StringIndexerInverse`

# Answer 4

**Correct: C** — it reverses `StringIndexer`.

**Wrong:**

A: slices vectors.

B: not a Spark ML class.

D: not a class.

# Question 5

Which Spark ML transformer splits a text column into words?

A. `PCA`

B. `Tokenizer`

C. `Bucketizer`

D. `Imputer`

# Answer 5

**Correct:** B — it lowercases and splits on whitespace.

**Wrong:**

A: reduces dimensions.

C: bins numbers.

D: fills missing values.

# Question 6

Which pair of Spark ML stages builds TF-IDF features from tokenized text?

A. `Bucketizer` followed by `Imputer`
B. `VectorAssembler` followed by `KMeans`
C. `HashingTF` (or `CountVectorizer`) followed by `IDF`
D. `StringIndexer` followed by `PCA`

# Answer 6

**Correct: C** — term frequencies are rescaled by inverse document frequency.

**Wrong:**

A: not TF-IDF.

B: not TF-IDF.

D: not TF-IDF.

# Question 7

Which Spark ML transformer scales each row (sample) to unit norm?

A. `StandardScaler`

B. `Bucketizer`

C. `MinMaxScaler`

D. `Normalizer`

# Answer 7

**Correct: D** — it normalizes per row, unlike scalers that work per column.

**Wrong:**

A: scales per feature.

B: bins values.

C: scales per feature.

# Question 8

Which scaler preserves sparsity by dividing each feature by its maximum absolute value?

A. `OneHotEncoder`
B. `MaxAbsScaler`
C. `Imputer`
D. `StandardScaler` with centering

# Answer 8

**Correct:** B — zeros stay zero.

**Wrong:**

A: encodes categories.

C: fills missing values.

D: centering destroys sparsity.

# Question 9

What does `RFormula` do in Spark ML?

A. Computes ROC AUC

B. Registers a model

C. Builds a features vector and label column from an R-style formula such as `y ~ a + b`

D. Reads Delta tables

# Answer 9

**Correct: C** — it handles indexing and encoding for formula terms.

**Wrong:**

A: that is an evaluator.

B: not its role.

D: not its role.

# Question 10

Which transformer lets you apply a SQL statement to a DataFrame inside a pipeline using `__THIS__`?

A. `RFormula`

B. `Bucketizer`

C. `VectorAssembler`

D. `SQLTransformer`

# Answer 10

**Correct: D** — `__THIS__` stands for the input DataFrame.

**Wrong:**

- A: uses an R formula.
- B: bins numbers.
- C: combines columns.

# Question 11

Which transformer converts a numeric column to 0.0 or 1.0 using a threshold?

A. `PCA`
B. `Binarizer`
C. `Normalizer`
D. `IndexToString`

# Answer 11

**Correct:** B — values above the threshold become 1.0.

**Wrong:**

A: reduces dimensions.

C: scales rows.

D: maps indices to labels.

# Question 12

Which Spark ML component helps select the most relevant features using statistical tests?

A. `KMeans`

B. `Tokenizer`

C. `IndexToString`

D. `UnivariateFeatureSelector` (or `ChiSqSelector`)

# Answer 12

**Correct: D** — they score features against the label.

**Wrong:**

A: clusters rows.

B: splits text.

C: maps indices.

# Question 13

What does `PCA(k=3, inputCol="features", outputCol="pca")` produce after fit and transform?

A. A dataset with 3 rows

B. A model with 3 trees

C. A column with the first 3 principal components

D. A column with 3 clusters

# Answer 13

**Correct: C** — k sets the number of components.

**Wrong:**

A: not PCA.

B: not PCA.

D: that is KMeans.

# Question 14

Why should `VectorAssembler` usually be one of the last feature stages in a pipeline?

A. It must run before any imputation

B. It registers the pipeline

C. It combines the already processed columns into the single features vector the model needs

D. It trains the model

# Answer 14

**Correct: C** — upstream stages create the processed columns first.

**Wrong:**

A: it needs processed columns.

B: not its role.

D: it is a transformer.

# Question 15

Which call saves a matplotlib figure to a run as an image artifact?

A. `mlflow.log_metric(fig)`

B. `mlflow.set_figure(fig)`

C. `mlflow.log_param(fig)`

D. `mlflow.log_figure(fig, "plot.png")`

# Answer 15

**Correct: D** — log_figure handles matplotlib and plotly figures.

**Wrong:**

A: metrics are numbers.

B: not an API.

C: params are small values.

# Question 16

Which call saves a Python dictionary as a JSON file artifact in a run?

A. `mlflow.log_dict(d, "config.json")`

B. `mlflow.log_params_file(d)`

C. `mlflow.set_dict(d)`

D. `mlflow.log_tag(d)`

# Answer 16

**Correct: A** — the dict is stored as an artifact file.

**Wrong:**

B: not an API.

C: not an API.

D: not an API.

# Question 17

Which call stores a string of text as an artifact file?

A. `mlflow.log_text(text, "notes.txt")`

B. `mlflow.log_metric(text)`

C. `mlflow.log_string(text)`

D. `mlflow.set_tag("notes.txt", file)`

# Answer 17

**Correct: A** — log_text writes a text artifact.

**Wrong:**

B: metrics are numbers.

C: not an API.

D: tags are not files.

# Question 18

A filter string needs runs with `max_depth` equal to 5. Params are stored as strings. Which `filter_string` is correct?

A. `"max_depth:5"`

B. `"params.max_depth = '5'"`

C. `"metrics.max_depth = 5"`

D. `"tags.max_depth == 5"`

# Answer 18

**Correct:** B — param values are compared as quoted strings.

**Wrong:**

A: not valid syntax.

C: max_depth is a param, not a metric.

D: wrong namespace and operator.

# Question 19

Which filter string finds runs with RMSE below 3?

A. `"tags.rmse < 3"`

B. `"metrics.rmse < 3"`

C. `"rmse = 3"`

D. `"params.rmse < 3"`

# Answer 19

**Correct: B** — metrics are compared numerically.

**Wrong:**

A: tags are strings.

C: namespace missing and not a less-than.

D: rmse is a metric.

# Question 20

Which call returns the full history of a metric such as loss over steps for a run?

A. `mlflow.log_metric_history()`

B. `mlflow.get_metric(run_id)`

C. `mlflow.search_runs(run_id)`

D. `MlflowClient().get_metric_history(run_id, "loss")`

# Answer 20

**Correct: D** — it returns all logged values with steps.

**Wrong:**

A: not an API.

B: not an API.

C: searches runs, not step history.

# Question 21

Which call downloads a run's artifacts to local storage?

A. `mlflow.artifacts.download_artifacts(run_id=..., artifact_path=...)`

B. `mlflow.get_files()`

C. `mlflow.pull_artifacts()`

D. `mlflow.load_run_dir()`

# Answer 21

**Correct: A** — it copies artifacts locally.

**Wrong:**

B: not an API.

C: not an API.

D: not an API.

# Question 22

Which call lists experiments in the tracking server?

A. `mlflow.search_runs()`
B. `mlflow.search_experiments()`
C. `mlflow.get_experiments_df()`
D. `mlflow.list_models()`

# Answer 22

**Correct:** **B** — it returns experiment objects.

**Wrong:**

A: lists runs.

C: not an API.

D: not an API.

# Question 23

Which call gets an experiment object by its path or name?

A. `mlflow.get_name(experiment)`
B. `mlflow.find_experiment(name)`
C. `mlflow.search_runs(name)`
D. `mlflow.get_experiment_by_name(name)`

# Answer 23

**Correct: D** — it returns the experiment or None.

**Wrong:**

A: not an API.

B: not an API.

C: searches runs.

# Question 24

What does `mlflow.log_input(dataset)` help record?

A. The GPU type

B. The dataset used with a run, for lineage

C. The alias

D. The model signature

# Answer 24

**Correct:** B — it ties data to the run.

**Wrong:**

A: separate.

C: separate.

D: separate.

# Question 25

What is stored in a run's `params` as opposed to its `metrics`?

A. They are identical

B. Params are string inputs such as hyperparameters; metrics are numeric results

C. Params hold images; metrics hold text

D. Params are numeric results only

# Answer 25

**Correct: B** — they serve different roles.

**Wrong:**

A: they differ.

C: not true.

D: that is metrics.

# Question 26

Which call deletes a run from tracking?

A. `mlflow.clear()`

B. `mlflow.end_run(run_id)`

C. `mlflow.remove_metric(run_id)`

D. `mlflow.delete_run(run_id)`

# Answer 26

**Correct: D** — it moves the run to deleted state.

**Wrong:**

A: not an API.

B: ends the active run.

C: not an API.

# Question 27

What does Delta `CLONE` do?

A. Creates a copy of a table (shallow clones reference the source data files)

B. Deletes a table

C. Encrypts the table

D. Trains a model

# Answer 27

**Correct: A** — clones are useful for testing and experiments.

**Wrong:**

B: not what it does.

C: not what it does.

D: not what it does.

# Question 28

What does Delta Change Data Feed provide?

A. Row-level changes (inserts, updates, deletes) between table versions
B. Faster clusters
C. Automatic feature selection
D. Model serving

# Answer 28

**Correct: A** — downstream jobs can consume changes only.

**Wrong:**

B: not its purpose.

C: not its purpose.

D: not its purpose.

# Question 29

Which setting is required for a Structured Streaming write so the query can recover after restart?

A. An MLflow run

B. A GPU cluster

C. A model alias

D. A checkpoint location

# Answer 29

**Correct: D** — checkpoints store progress.

**Wrong:**

A: not required.

B: not required.

C: not required.

# Question 30

What does `trigger(availableNow=True)` do in a streaming query?

A. Runs forever with no batches
B. Processes all available data and then stops
C. Deletes the checkpoint
D. Re-reads the whole table each second

# Answer 30

**Correct:** **B** — it behaves like an incremental batch.

**Wrong:**

A: not the behavior.

C: not the behavior.

D: not the behavior.

# Question 31

Which call reads a Delta table as a stream?

A. `spark.table("t").stream()`

B. `spark.read.stream("t")`

C. `spark.streamTable("t")`

D. `spark.readStream.table("catalog.schema.t")`

# Answer 31

**Correct: D** — readStream creates a streaming DataFrame.

**Wrong:**

A: not valid.

B: not valid.

C: not valid.

# Question 32

Which streaming sink option lets you run batch logic such as scoring on each micro-batch?

A. `foreachBatch`

B. `mode("overwrite")`

C. `foreachPartition` on the driver

D. `coalesce(1)`

# Answer 32

**Correct: A** — it exposes each micro-batch as a DataFrame.

**Wrong:**

B: not a streaming hook.

C: not the micro-batch hook.

D: not a streaming hook.

# Question 33

Which output mode writes only new rows added since the last trigger?

A. Overwrite
B. Complete
C. Append
D. Update

# Answer 33

**Correct: C** — append emits new rows only.

**Wrong:**

A: not a streaming output mode name.

B: rewrites the whole result.

D: emits changed rows.

# Question 34

A feature pipeline must refresh a feature table whenever new rows land in a source Delta table. Which approach fits?

A. Manually edit the feature table

B. Recreate the cluster

C. A streaming or scheduled job that computes features and writes them with `mode="merge"`

D. Overwrite the model

# Answer 34

**Correct: C** — merging by key keeps features current.

**Wrong:**

A: error prone.

B: not related.

D: not the feature pipeline.

# Question 35

Which Delta constraint rejects rows that violate a rule such as age >= 0?

A. A ZORDER constraint

B. An OPTIMIZE rule

C. A CHECK constraint

D. A VACUUM rule

# Answer 35

**Correct: C** — writes that violate it fail.

**Wrong:**

A: not a constraint.

B: not a constraint.

D: not a constraint.

# Question 36

Why does Delta's ACID support matter for ML pipelines?

A. It makes models more accurate

B. It disables time travel

C. Concurrent reads and writes see consistent table versions

D. It removes the need for validation

# Answer 36

**Correct: C** — training does not read half-written data.

**Wrong:**

A: not implied.

B: it enables time travel.

D: not implied.

# Question 37

A training dataset must be frozen for audit. Which approach is simplest in Delta?

A. Record the table version and read it with time travel, or clone the table

B. Delete the table afterward

C. Rely on the latest version

D. Cache it in a notebook

# Answer 37

**Correct: A** — both preserve the exact data.

**Wrong:**

B: loses data.

C: changes over time.

D: not durable.

# Question 38

Which write pattern avoids duplicate feature rows when reprocessing the same data?

A. Overwrite with an empty DataFrame
B. Append every time
C. Drop the primary key
D. Merge on the primary key

# Answer 38

**Correct: D** — upserts replace instead of duplicating.

**Wrong:**

A: deletes data.

B: creates duplicates.

C: removes uniqueness.

# Question 39

A feature is right-skewed. Which statement about mean and median is typical?

A. The mean is larger than the median

B. The mean is smaller than the median

C. They are always equal

D. The median is undefined

# Answer 39

**Correct: A** — the long right tail pulls the mean up.

**Wrong:**

B: typical of left skew.

C: only for symmetric data.

D: it is defined.

# Question 40

A value is 70, the mean is 50, and the standard deviation is 10. What is its z-score?

A. 2.0

B. 0.2

C. 7.0

D. 20

# Answer 40

**Correct: A** — (70 - 50) / 10.

**Wrong:**

B: inverted.

C: incorrect.

D: forgets to divide.

# Question 41

Min-max scaling a value of 30 to the range 0 to 1 when the minimum is 10 and the maximum is 50 gives:

A. 0.5

B. 0.6

C. 0.3

D. 1.0

# Answer 41

**Correct: A** — (30 - 10) / (50 - 10).

**Wrong:**

B: incorrect.

C: incorrect.

D: that would be 50.

# Question 42

What does a box plot show?

A. The confusion matrix
B. Model coefficients
C. Median, quartiles, and potential outliers
D. Only the mean

# Answer 42

**Correct: C** — the box spans Q1 to Q3.

**Wrong:**

A: not shown.

B: not shown.

D: not just the mean.

# Question 43

Which correlation measure captures monotonic but non-linear relationships?

A. R2 only
B. Variance inflation factor
C. Spearman rank correlation
D. Pearson correlation

# Answer 43

**Correct: C** — it uses ranks.

**Wrong:**

A: not a correlation measure.

B: detects collinearity.

D: measures linear association.

# Question 44

Correlation between two features is high. What can you conclude?

A. They are identical columns
B. They must both be dropped
C. They are associated, but this does not prove causation
D. One causes the other

# Answer 44

**Correct: C** — association is not causation.

**Wrong:**

A: not implied.

B: not required.

D: not proven.

# Question 45

A feature has the same value in every row. What should you usually do?

A. Drop it because it carries no information
B. One-hot encode it
C. Use it as the key
D. Keep it as the label

# Answer 45

**Correct: A** — zero variance features add nothing.

**Wrong:**

B: adds nothing.

C: not appropriate.

D: not appropriate.

# Question 46

Which chart best shows the distribution of a single numeric feature?

A. A Sankey chart

B. A pie chart of rows

C. A treemap

D. A histogram

# Answer 46

**Correct: D** — it shows frequencies across bins.

**Wrong:**

A: not a distribution.

B: not a distribution.

C: not a distribution.

# Question 47

A scatter plot of a feature against the target is used to:

A. Reveal relationships, trends, and outliers
B. Train the model
C. Deploy the endpoint
D. Register the model

# Answer 47

**Correct: A** — it is a basic EDA tool.

**Wrong:**

B: not its purpose.

C: not its purpose.

D: not its purpose.

# Question 48

A sample used for EDA must be representative. Which practice helps most?

A. Using only rows with no nulls
B. Random or stratified sampling
C. Using only the newest rows
D. Taking the first 1000 rows of a sorted table

# Answer 48

**Correct:** **B** — it avoids bias in the sample.

**Wrong:**

A: can be biased.

C: can be biased.

D: can be biased.

# Question 49

What does a Naive Bayes classifier assume?

A. Features are conditionally independent given the class
B. Features are perfectly correlated
C. Labels are continuous
D. Data is time ordered

# Answer 49

**Correct: A** — this simplifies probability estimation.

**Wrong:**

B: opposite.

C: classification uses classes.

D: not assumed.

# Question 50

What does a support vector machine try to maximize?

A. The margin between classes

B. The learning rate

C. The number of trees

D. The number of clusters

# Answer 50

**Correct: A** — it finds the separating boundary with the widest margin.

**Wrong:**

B: not SVM.

C: not SVM.

D: not SVM.

# Question 51

Which criteria are commonly used by decision trees to choose splits for classification?

A. Silhouette or elbow
B. VIF or PCA
C. ROC AUC or R2
D. Gini impurity or entropy

# Answer 51

**Correct: D** — they measure node impurity.

**Wrong:**

A: clustering diagnostics.

B: not split criteria.

C: evaluation metrics.

# Question 52

What function maps a linear score to a probability in logistic regression?

A. ReLU

B. The identity function

C. Min-max scaling

D. The sigmoid (logistic) function

# Answer 52

**Correct: D** — it outputs values between 0 and 1.

**Wrong:**

A: not used for probabilities here.

B: would not bound outputs.

C: not the link function.

# Question 53

Which loss is typically optimized by logistic regression?

A. MAPE

B. Log loss (cross-entropy)

C. Hinge on distances only

D. Silhouette

# Answer 53

**Correct:** B — it penalizes wrong probabilities.

**Wrong:**

A: regression metric.

C: not logistic regression's loss.

D: clustering.

# Question 54

What does k-means require you to specify in advance?

A. The tree depth

B. The learning rate only

C. The labels

D. The number of clusters k

# Answer 54

**Correct: D** — k is a hyperparameter.

**Wrong:**

A: not used.

B: not required.

C: k-means is unsupervised.

# Question 55

Why is feature scaling important before k-means?

A. It uses labels

B. It uses trees

C. It cannot handle numbers

D. It uses distances, so large-range features would dominate

# Answer 55

**Correct: D** — scaling balances feature influence.

**Wrong:**

A: it does not.

B: it does not.

C: it can.

# Question 56

In bagging, how is each tree trained?

A. On the previous tree's residuals

B. On a single row

C. On the test data

D. On a bootstrap sample of the training data

# Answer 56

**Correct: D** — random resampling introduces diversity.

**Wrong:**

A: that is boosting.

B: not bagging.

C: invalid.

# Question 57

What is a typical trade-off between learning rate and number of trees in gradient boosting?

A. They are unrelated
B. A smaller rate needs fewer trees
C. A larger rate always generalizes better
D. A smaller learning rate usually needs more trees

# Answer 57

**Correct: D** — each tree contributes less.

**Wrong:**

A: they interact.

B: opposite.

C: can overfit.

# Question 58

Which technique creates synthetic minority examples to address imbalance?

A. SMOTE

B. Bucketizer

C. VIF

D. PCA

# Answer 58

**Correct: A** — apply it to training data only.

**Wrong:**

B: binning.

C: multicollinearity measure.

D: dimension reduction.

# Question 59

Where should oversampling be applied to avoid leakage?

A. Only on the training data after the split
B. On the full dataset before splitting
C. On the labels only
D. On the test set only

# Answer 59

**Correct: A** — test data must remain untouched.

**Wrong:**

B: leaks synthetic neighbors.

C: invalid.

D: invalid.

# Question 60

What is a dummy baseline model used for?

A. Deploying the model
B. Replacing validation
C. Providing a minimum performance level to beat
D. Reducing leakage

# Answer 60

**Correct: C** — it shows whether the model adds value.

**Wrong:**

A: not its purpose.

B: not its purpose.

D: not its purpose.

# Question 61

A tree-based importance score can be biased toward which kind of feature?

A. High-cardinality or continuous features

B. Constant features

C. Binary features

D. Label columns only

# Answer 61

**Correct: A** — they offer many split points.

**Wrong:**

B: have no splits.

C: less likely to be biased upward.

D: not relevant.

# Question 62

What is the main purpose of a train/validation/test split?

A. Only the test set is used
B. Train fits the model, validation tunes it, and test gives a final unbiased estimate
C. All three are used to fit weights
D. They are shuffled together at inference

# Answer 62

**Correct:** B — each split has a distinct role.

**Wrong:**

A: need training data.

C: validation and test are held out.

D: not a purpose.

# Question 63

In linear regression, what do residuals represent?

A. The regularization terms

B. The learning rates

C. The differences between actual and predicted values

D. The feature means

# Answer 63

**Correct: C** — patterns in residuals indicate model problems.

**Wrong:**

A: not residuals.

B: not residuals.

D: not residuals.

# Question 64

Which statement about stacking is true?

A. It only works for clustering

B. A meta-model learns to combine predictions of base models

C. It trains one tree on all labels

D. It removes the need for validation

# Answer 64

**Correct:** B — it is an ensemble technique.

**Wrong:**

A: not true.

C: not stacking.

D: not true.

# Question 65

A dataset does not fit in the memory of one machine and you need to train a model. Which choice scales out?

A. `df.toPandas()` then fit

B. scikit-learn on the driver

C. Increasing `maxIter` only

D. Spark ML or a distributed library such as distributed XGBoost

# Answer 65

**Correct: D** — they train across workers.

**Wrong:**

A: collects to the driver.

B: limited to one machine.

C: does not fix memory.

# Question 66

What is data parallelism in distributed training?

A. Workers train different unrelated models

B. Each worker trains on a shard of data and gradients or updates are synchronized

C. Only the driver trains

D. Each worker holds a different dataset type

# Answer 66

**Correct: B** — the model is replicated across workers.

**Wrong:**

A: that is a different pattern.

C: not parallel.

D: not the definition.

# Question 67

Which Databricks-supported tools distribute deep learning training across GPUs or nodes?

A. `VACUUM`

B. `TorchDistributor` or Horovod

C. `OneHotEncoder`

D. `dbutils.fs`

# Answer 67

**Correct:** B — they coordinate distributed training.

**Wrong:**

A: Delta maintenance.

C: feature encoder.

D: file utility.

# Question 68

A team wants to run many small independent model fits (one per product) quickly. Which is a good approach?

A. Group by product and train each model in parallel with `applyInPandas`

B. One huge model on the driver for all products

C. Use `coalesce(1)`

D. Collect each product to the driver sequentially

# Answer 68

**Correct: A** — each group trains on a worker.

**Wrong:**

B: may not fit the need.

C: removes parallelism.

D: slow.

# Question 69

Which action is a safe way to explore a huge DataFrame in pandas?

A. Call `collect()` on everything

B. Take a random sample with `df.sample(fraction)` and then `toPandas()`

C. Call `repartition(1)`

D. Call `toPandas()` on the full DataFrame

# Answer 69

**Correct:** B — sampling avoids driver overflow.

**Wrong:**

A: may crash the driver.

C: does not reduce data.

D: may crash the driver.

# Question 70

Which scaling approach helps when Spark joins create too many small partitions or very large shuffles?

A. Convert everything to CSV

B. Tune partitioning and use broadcast joins for small tables

C. Use more registered models

D. Disable caching

# Answer 70

**Correct:** B — this reduces shuffle cost.

**Wrong:**

A: not helpful.

C: not related.

D: not helpful.

# Question 71

What does a Single Node cluster lack compared to a multi-node cluster?

A. MLflow support
B. Notebook support
C. Worker nodes for distributing Spark tasks
D. A Python runtime

# Answer 71

**Correct: C** — only the driver is used.

**Wrong:**

A: it supports MLflow.

B: it supports notebooks.

D: it has Python.

# Question 72

Which resource is most important for running `toPandas()` on a moderately large result?

A. Model alias count

B. Git folder size

C. Driver memory

D. Number of workers

# Answer 72

**Correct: C** — the data is collected to the driver.

**Wrong:**

A: unrelated.

B: unrelated.

D: does not hold collected data.

# Question 73

Which `fmin` argument stops the search after a number of seconds?

A. `max_depth`

B. `timeout`

C. `parallelism`

D. `numFolds`

# Answer 73

**Correct:** B — it bounds total search time.

**Wrong:**

A: a model hyperparameter.

C: SparkTrials argument.

D: Spark ML CV.

# Question 74

Which feature lets Hyperopt stop when the loss has not improved for several trials?

A. `STATUS_FAIL`
B. `space_eval`
C. `hp.choice` only
D. An early stopping function such as `no_progress_loss`

# Answer 74

**Correct: D** — passed as `early_stop_fn`.

**Wrong:**

A: marks a failed trial.

B: decodes parameters.

C: defines a space.

# Question 75

How can you make a Hyperopt search reproducible?

A. Disable MLflow

B. Increase `max_evals`

C. Pass a seeded random state through `rstate` (for example `np.random.default_rng(42)`)

D. Use more workers

# Answer 75

**Correct: C** — the seed fixes sampling.

**Wrong:**

A: does not fix randomness.

B: does not fix randomness.

D: does not fix randomness.

# Question 76

How can a single Hyperopt search compare different model types such as RandomForest and GradientBoosting?

A. Use `hp.choice` with nested search spaces for each model type

B. Run `fmin` on the test set

C. Use `hp.uniform` only

D. Use `df.randomSplit`

# Answer 76

**Correct: A** — the choice selects which sub-space is sampled.

**Wrong:**

B: invalid.

C: cannot pick model types.

D: unrelated.

# Question 77

What does returning `{"loss": None, "status": STATUS_FAIL}` from an objective indicate?

A. That the trial was best

B. That the model is registered

C. That the trial failed

D. That tuning is complete

# Answer 77

**Correct: C** — the status marks the trial outcome.

**Wrong:**

A: no.

B: no.

D: no.

# Question 78

Which method of Hyperopt `Trials` gives the dictionary describing the best trial?

A. `trials.winner`
B. `trials.top()`
C. `trials.predict()`
D. `trials.best_trial`

# Answer 78

**Correct: D** — it holds the result and misc info for the best trial.

**Wrong:**

A: not an attribute.

B: not an attribute.

C: not an attribute.

# Question 79

What are the axes of an ROC curve?

A. Loss versus epoch

B. True positive rate versus false positive rate

C. RMSE versus R2

D. Precision versus recall

# Answer 79

**Correct: B** — points are computed across thresholds.

**Wrong:**

A: a learning curve.

C: not an ROC curve.

D: that is the PR curve.

# Question 80

What does an AUC of 0.5 indicate?

A. A perfect classifier

B. Performance no better than random guessing

C. A perfectly wrong classifier

D. A calibrated model

# Answer 80

**Correct: B** — the ROC curve follows the diagonal.

**Wrong:**

A: that is 1.0.

C: that would be 0.0.

D: not implied.

# Question 81

What is specificity?

A. TN / (TN + FP)
B. TP / (TP + FN)
C. TP / (TP + FP)
D. FP / (FP + TN)

# Answer 81

**Correct: A** — the true negative rate.

**Wrong:**

B: that is recall.

C: that is precision.

D: that is the false positive rate.

# Question 82

What does balanced accuracy compute?

A. The sum of all errors
B. The AUC
C. The maximum class accuracy
D. The average of recall across classes

# Answer 82

**Correct: D** — it corrects for class imbalance.

**Wrong:**

A: not its definition.

B: different metric.

C: not its definition.

# Question 83

When is RMSLE preferred over RMSE?

A. When targets are negative
B. When relative errors matter and targets span orders of magnitude
C. When classifying images
D. When labels are binary

# Answer 83

**Correct:** B — it uses logs of target values.

**Wrong:**

A: log is not defined.

C: not a classification metric.

D: not a classification metric.

# Question 84

For a dataset with 2 percent positives, what is the baseline of the precision-recall curve (random classifier precision)?

A. 0.98

B. 0.5

C. About 0.02

D. 1.0

# Answer 84

**Correct: C** — random precision equals the positive prevalence.

**Wrong:**

A: that is the negative rate.

B: that is for ROC AUC.

D: that is perfect.

# Question 85

A missed fraud case costs much more than a false alarm. How should the decision threshold be set?

A. Remove the threshold

B. Raise the threshold to maximize precision only

C. Lower the threshold to favor recall, guided by the business costs

D. Use 0.5 always

# Answer 85

**Correct: C** — cost-sensitive thresholding aligns the model with costs.

**Wrong:**

A: needed for labels.

B: misses more fraud.

D: ignores costs.

# Question 86

10-fold cross-validation trains each model on what share of the data?

A. 50 percent

B. 90 percent

C. 10 percent

D. 100 percent

# Answer 86

**Correct: B** — 9 of 10 folds are used for training.

**Wrong:**

A: incorrect.

C: that is the validation fold.

D: would include validation data.

# Question 87

Which Lakehouse Monitoring profile type fits a table of model inputs, predictions, and labels?

A. Delta profile

B. Time series profile only

C. Inference profile

D. Snapshot profile

# Answer 87

**Correct: C** — it is designed for ML inference logs.

**Wrong:**

A: not a profile type.

B: does not include model quality.

D: is for non-time-series tables.

# Question 88

What does Lakehouse Monitoring create when a monitor is set up?

A. A new Git repo

B. A new cluster policy

C. A new model alias

D. Metric tables (profile and drift) and an auto-generated dashboard

# Answer 88

**Correct: D** — they summarize statistics and drift.

**Wrong:**

A: not created.

B: not created.

C: not created.

# Question 89

What is a baseline table used for in drift monitoring?

A. Comparing current data against reference data such as training data
B. Defining aliases
C. Logging job failures
D. Storing model weights

# Answer 89

**Correct: A** — drift is measured relative to the baseline.

**Wrong:**

B: not its purpose.

C: not its purpose.

D: not its purpose.

# Question 90

A monitor flags a large drift in a key feature. What is a sensible next step?

A. Delete the monitor

B. Increase the endpoint size

C. Investigate the cause and consider retraining or fixing the upstream data

D. Ignore it

# Answer 90

**Correct: C** — drift may be real change or a pipeline bug.

**Wrong:**

A: removes visibility.

B: does not fix drift.

D: may degrade quality.

# Question 91

What is a canary deployment?

A. Deleting the old model

B. Training on production traffic only

C. Deploying only on weekends

D. Sending a small share of traffic to a new version before full rollout

# Answer 91

**Correct: D** — it limits the blast radius of problems.

**Wrong:**

A: not canary.

B: not canary.

C: not canary.

# Question 92

What is shadow deployment?

A. Disabling the old model

B. Running a new model on live traffic without using its predictions for decisions

C. Training offline only

D. Serving the new model to all users

# Answer 92

**Correct:** **B** — its outputs are logged and compared.

**Wrong:**

A: not shadow.

C: not shadow.

D: not shadow.

# Question 93

A custom pyfunc model needs to load a vectorizer file once at startup. Which method is designed for this?

A. `__del__`

B. `start_run()`

C. `load_context(self, context)`

D. `predict()` on every call

# Answer 93

**Correct: C** — it runs when the model is loaded.

**Wrong:**

A: not designed for loading.

B: not a pyfunc method.

D: reloading is wasteful.

# Question 94

Which artifact do you pass to a pyfunc model so it can access extra files at load time?

A. `tags={...}`

B. `artifacts={...}` in `mlflow.pyfunc.log_model`

C. `conda_env={...}`

D. `signature={...}`

# Answer 94

**Correct:** B — they are available through `context.artifacts`.

**Wrong:**

A: labels the model.

C: defines the environment only.

D: defines schema.

# Question 95

A model endpoint is returning errors after a library update in the training notebook. What is the most likely issue?

A. The logged model's dependencies do not match the serving environment or code expectations

B. The feature table is a Delta table

C. The endpoint has too many aliases

D. The experiment name changed

# Answer 95

**Correct: A** — dependencies should be captured with the model.

**Wrong:**

B: not the cause.

C: not related.

D: not related.

# Question 96

Which option reduces cluster start-up time for frequent jobs?

A. Turning off MLflow
B. Running `VACUUM` more often
C. Instance pools or serverless compute
D. Increasing `maxDepth`

# Answer 96

**Correct: C** — pools keep warm instances ready.

**Wrong:**

A: not related.

B: not related.

D: a model hyperparameter.

# Question 97

Which command exits a notebook and returns a value to the caller (for example in a job or `notebook.run`)?

A. `dbutils.notebook.exit(value)`

B. `dbutils.fs.rm(value)`

C. `mlflow.end_run(value)`

D. `spark.stop(value)`

# Answer 97

**Correct: A** — it ends the notebook with a result string.

**Wrong:**

B: deletes files.

C: ends a run.

D: stops Spark.

# Question 98

Which cell magic runs SQL in a Python notebook?

A. `%scala` only

B. `%sql`

C. `%md`

D. `%sh`

# Answer 98

**Correct:** **B** — it switches the cell language to SQL.

**Wrong:**

A: Scala.

C: markdown.

D: shell.

# Question 99

Which permission setting controls who can view or edit an MLflow experiment?

A. Delta constraints
B. Notebook themes
C. Cluster policies
D. Experiment permissions (ACLs)

# Answer 99

**Correct: D** — access is managed per experiment.

**Wrong:**

A: limit data values.

B: cosmetic.

C: limit configs.

# Question 100

Which feature lets teams share governed, reusable SQL functions and models under the same catalog namespace as tables?

A. DBFS root

B. Cluster init scripts

C. Notebook revision history

D. Unity Catalog schemas that hold tables, functions, volumes, and models

# Answer 100

**Correct: D** — objects live in the same three-level namespace.

**Wrong:**

A: ungoverned storage.

B: not data governance.

C: not governance.